

Vocabulary

Parent function: the simplest member of a family of functions — no shifts, stretches, or reflections applied.

Domain: the set of all allowed input (x) values a function can use.

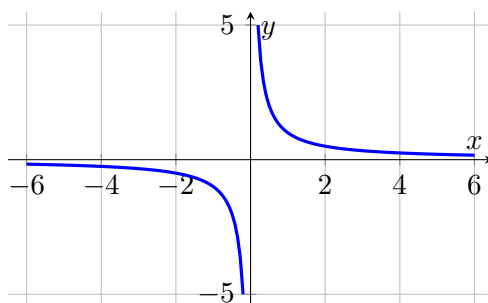
Range: the set of all output (y) values a function actually produces.

We do together. Identify the parent function of $f(x) = \frac{1}{x}$ and give its domain and range.

Step 1. The rule is a fraction with x in the denominator, so this is the reciprocal parent function.

Step 2. A denominator can never equal 0, so x cannot be 0: domain is $x \neq 0$.

Step 3. Since $y = 1/x$ never produces an output of exactly 0 either, the range is $y \neq 0$.



$y = 1/x$ — two branches, never touching either axis (domain $x \neq 0$, range $y \neq 0$).

You Try. Solve each. Show every step, then name the answer.

a) Identify the parent function of $f(x) = x^3$ and give its domain and range. cubic; domain all real numbers, range all real numbers

b) Identify the parent function of $f(x) = |x|$ and give its range. absolute value; range $y \geq 0$

Summary (fill in): To identify a parent function: match the equation to one of the 6 known shapes, then check for restrictions caused by a root or a denominator.